Taru made those notes ;)

Thoughts/future directions

- Conduct experiments to test if findings of this paper apply to different datasets/applications (multimodal datasets, non-linear networks, novel physics problems like phase transitions with high spectral variation)
    - start with some basic code to replicate this paper
    - generate/find datasets
- Vary initializations and study generalization behavior of NN – metrics to include Representational Similarity Matrix (hidden learning), Grokking related metrics, comment on spectral performance etc.
- Optional but interesting – expand to architectures with attention mechanism
- Our contribution:
    - Code
    - New applications / metrics
    - Observations for grokking, spectral bias and nonlinear networks (focus on experimentation instead of mathematical rigor)
    - Paper titles
        - *"Hidden Learning and Grokking in High Frequency and Multimodal Learning"*
        - *"Impact of absolute and relative initialization on spectral bias and grokking"*
        - *"The Lazy-Rich Transition in Nonlinear Neural Networks"*

The paper tracks 2 things:

- Network weights:
    - Initialization (experiments with)
    - Evolution during training (tracks)
- Task specific representations
    - Output of a layer (hidden or not) for a specific input.
    - How the hidden layer outputs change because of those weight changes. This is what the **Representational Similarity Matrix (RSM)** (tracks)
- Important: network in this paper is strictly linear — no activation function at all. *(Opportunity: replicating this study for non-linear functions can be a future direction)*
- Related papers / further reading:

Exact solutions to the nonlinear dynamics of learning in deep linear neural networks

OnLazy Training in Differentiable Programming

Get rich quick: exact solutions reveal how unbalanced initializations promote rapid feature learning

Lazy vs Rich Regime

Lazy Regime The network fits the training data by making minimal changes to its weights. It stays close to its starting point in weight space.

- Overparametrized NN, typical of infinitely wide architectures, large variance initialization

- Weights move very little from initialization
- Hidden layer representations barely change during training
- The network is essentially doing **kernel regression** — fitting a smooth function on top of fixed random features
- The NTK stays frozen — the network's "attention profile" never updates

(Opportunity: attention can be a future direction)

- Learning curves are **exponential** — smooth, fast, predictable
- This can be thought of as overfitting or lack of generalization
- Compute cost should be low, cost Is lack of generalization

Rich Regime The network fits the training data by **substantially reorganizing its weights**.

- Weights move substantially from initialization
- Weight initialization + differences in weights between layers (absolute and relative scale)
- Hidden layer representations actively evolve and develop task-specific structure
- The network discovers the true underlying structure of the problem
- The NTK evolves throughout training — the network's attention profile updates
- Learning curves are **sigmoidal** — long plateau, then sudden jump, then plateau again
- Strongest features are learned first, weaker ones later *(Opportunity: this may be the connection to spectral bias – though its an open /currently unresearched question)*

Because modern deep learning success likely comes more from rich feature learning than lazy (debatable / task specific)

Data used in the paper

- This paper is mostly theoretical, and its experiments are based on small/ synthetically generated data sources. It uses Gaussian synthetic inputs with analytically specified input-output correlations
- *(Opportunity: use larger data sets and observe results, images, natural language, physics data etc.)*

Initialization Approaches (pg. 21 appendix)

λ-balanced initializations

- family of initializations where layers are scaled relative to each other using a parameter λ

Further reading : Section4.pdf

Training / Architecture

- Architecture
  - full batch GD
  - MSE
  - input data is whitened, i.e. $\tilde{\Sigma}xx = I$
  - Initialization of weights (W1=W2 when Lambda = 0)

  - Hidden dimension of network is defined to prevent overparameterization
  - Funnel /Square/Anti-funnel networks

- Key statistics
    - Q = weight matrix = [W1 W2] ; metric QQ^T (t)
    - W2W1(t)
    - RSM (representational similarity matrices)
    - NTK

Findings

- Lambda around 0 results in highest hidden layer learning regardless of other aspects of the architecture
- Brief discussion about numerical stability, other network architectures, initialization appraoches